



A necessary shift: Many promising careers have been interrupted, or in some cases ended, by injuries that might have been avoided with better monitoring and targeted training. The adoption of sports genomics could change this dynamic by providing precise, actionable information on each athlete's unique strengths and vulnerabilities. GETTY IMAGES

without its challenges. It cannot guarantee that a genetically gifted athlete will succeed, nor can it predict exact outcomes. Training quality, mental resilience, access to facilities, and competitive experience all remain vital components of success. There are also ethical considerations. Genetic data is sensitive information, and its misuse could lead to discrimination or breaches of privacy. It is essential that any sports genomics programme in India is built on strict ethical guidelines, ensuring that data is stored securely, used only for the athlete's benefit, and never employed to exclude athletes based on genetic traits alone. Transparency and athlete consent must be at the core of any implementation.

For India to benefit from sports genomics, a clear and phased roadmap is required. The first step is awareness. Many coaches, administrators, and even sports scientists in the country are

unfamiliar with the capabilities and limitations of genomics. Education programmes, seminars, and pilot projects could help bridge this knowledge gap. The second step is capacity building. India must invest in training sports scientists, physiologists, and nutritionists to interpret and apply genetic data effectively. This would require partnerships between government agencies, national sports federations, private academies, and universities.

The financial cost of adopting sports genomics is modest compared to the price of preventable injuries or ineffective training investments. A single season-ending injury to a key player can cost teams and sponsors crores in lost opportunities. By contrast, the expense of genomic profiling is relatively low and provides benefits that can extend across an athlete's entire career. For sponsors and sports organisations, the value lies not only in protecting investments

but also in improving performance outcomes and career longevity.

The athletes themselves stand to gain the most. With a detailed understanding of their own bodies, they can take greater control over their training and recovery. They can understand why certain methods are recommended and how these align with their natural strengths. This clarity can build trust between athlete and coach, increase motivation, and create a stronger commitment to long-term goals.

If India adopts sports genomics decisively, it could reshape the country's sporting landscape within a decade. Athletes would face fewer interruptions from injuries, peak performance would be better timed for major events, and career spans could be extended. The science is ready, the evidence is growing, and global examples are abundant. What remains is the will to embrace this innovation and adapt it to the Indian context.